where d is the thickness of the plate and λ is the wavelength of the light beam. The phase retardation is the difference of the exponents in Equation (1.9-3) and is defined as

$$\Gamma = \frac{2\pi}{\lambda}(n_s - n_f)d \tag{1.9-4}$$

HALF-WAVE PLATE

A half-wave plate has a phase retardation of $\Gamma = \pi$. According, to Equation (1.9-4), an a-cut (surface is perpendicular to a axis) uniaxial plate will act as a half-wave plate provided the thickness is $d = \lambda/(2|n_e - n_o|)$. We will determine the effect of a half-wave plate on the polarization state of a transmitted light beam. The azimuth angle of the wave plate is taken as 45°, and the incident beam is vertically polarized. The Jones vector for the incident beam can be written

$$\mathbf{V} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{1.9-14}$$

and the Jones matrix for the half-wave plate is obtained by using Equations (1.9-9) to (1.9-11):

$$W = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \tag{1.9-15}$$

The Jones vector for the emerging beam is obtained by multiplying Equations (1.9-15) and (1.9-14); the result is

$$\mathbf{V}' = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i\begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{1.9-16}$$

This represents a beam of horizontally polarized light. The effect of the half-wave plate is to rotate the polarization by 90°. It can be shown that for a general azimuth angle ψ, the half-wave plate will rotate the polarization by an angle 2ψ (see Problem 1.19). In other words, linearly polarized light remains linearly polarized, except that the plane of polarization is rotated by an angle of 2ψ.

When the beam of incident light is circularly polarized, a half-wave plate will convert a beam of right-handed circularly polarized light into a beam of left-handed circularly polarized light and vice versa, regardless of the azimuth angle. The proof is left as an exercise (see Problem 1.19). Figure 1.17 illustrates the effect of a half-wave plate.

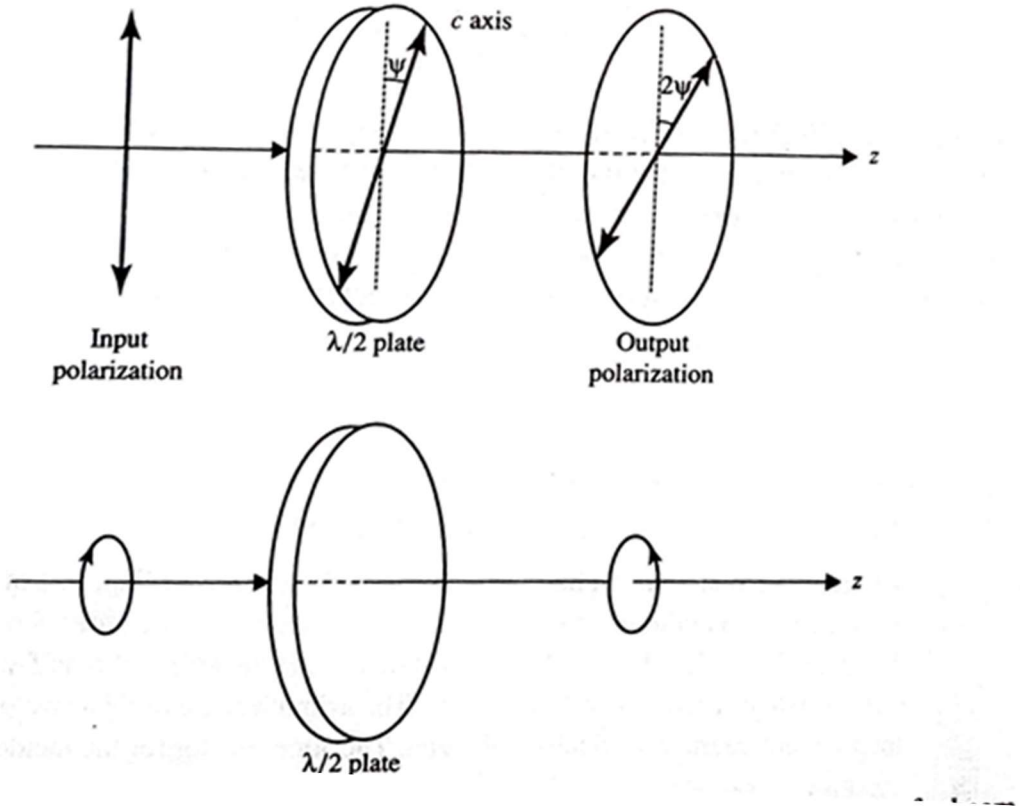


Fig: The effect of a half-wave plate on the polarization state of a beam

QUARTER-WAVE PLATE

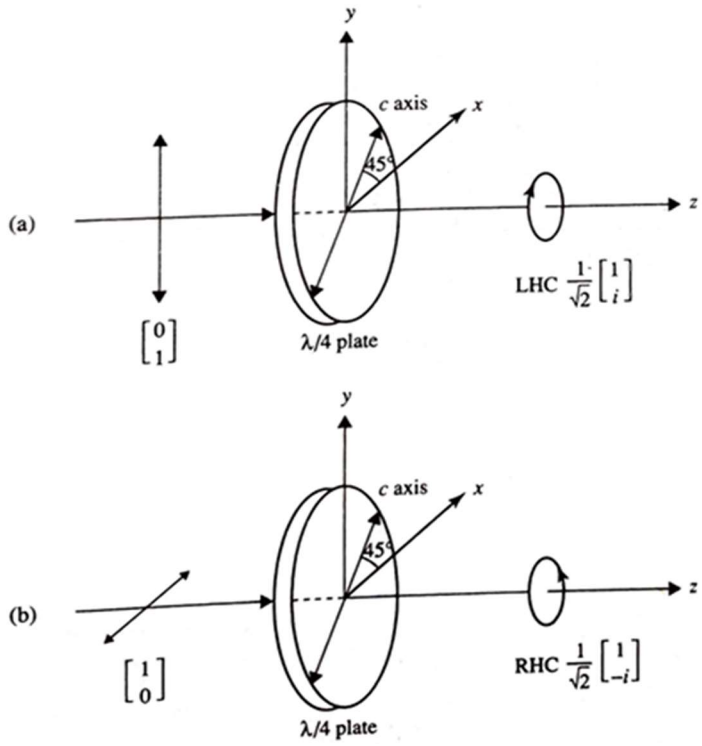
A quarter-wave plate has a phase retardation of $\Gamma = \pi/2$. If the plate is made of an a-cut (or b-cut) uniaxially anisotropic medium, the thickness is $d = \lambda/(4|n_e - n_o|)$ (or odd multiples thereof). Suppose again that the azimuth angle of the plate is $\psi = 45°$ and the incident beam is vertically polarized. The Jones vector for the incident beam is given again by Equation (1.9-14). The Jones matrix for this quarter-wave plate, according to Equation (1.9-11), is

$$W = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}e^{-i\pi/4} & 0\\ 0 & e^{i\pi/4}\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix} \tag{1.9-17}$$

The Jones vector for the emerging beam is obtained by multiplying Equations (1.9-17) and (1.9-14) and is given by

$$\mathbf{V}' = \frac{1}{\sqrt{2}}\begin{bmatrix}-i\\ 1\end{bmatrix} = \frac{-i}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} \tag{1.9-18}$$

This represents a beam of left-handed circularly polarized light. The effect of a 45° oriented quarter-wave plate is to convert a beam of vertically polarized light into a beam of left-handed circularly polarized light. If the incident beam is horizontally polarized, the emerging beam will be right-handed circularly polarized. In general, a quarter-wave plate can convert a linearly polarized light into an elliptically polarized light and vice versa. The effect of the quarter-wave plate is illustrated in Figure 1.18.



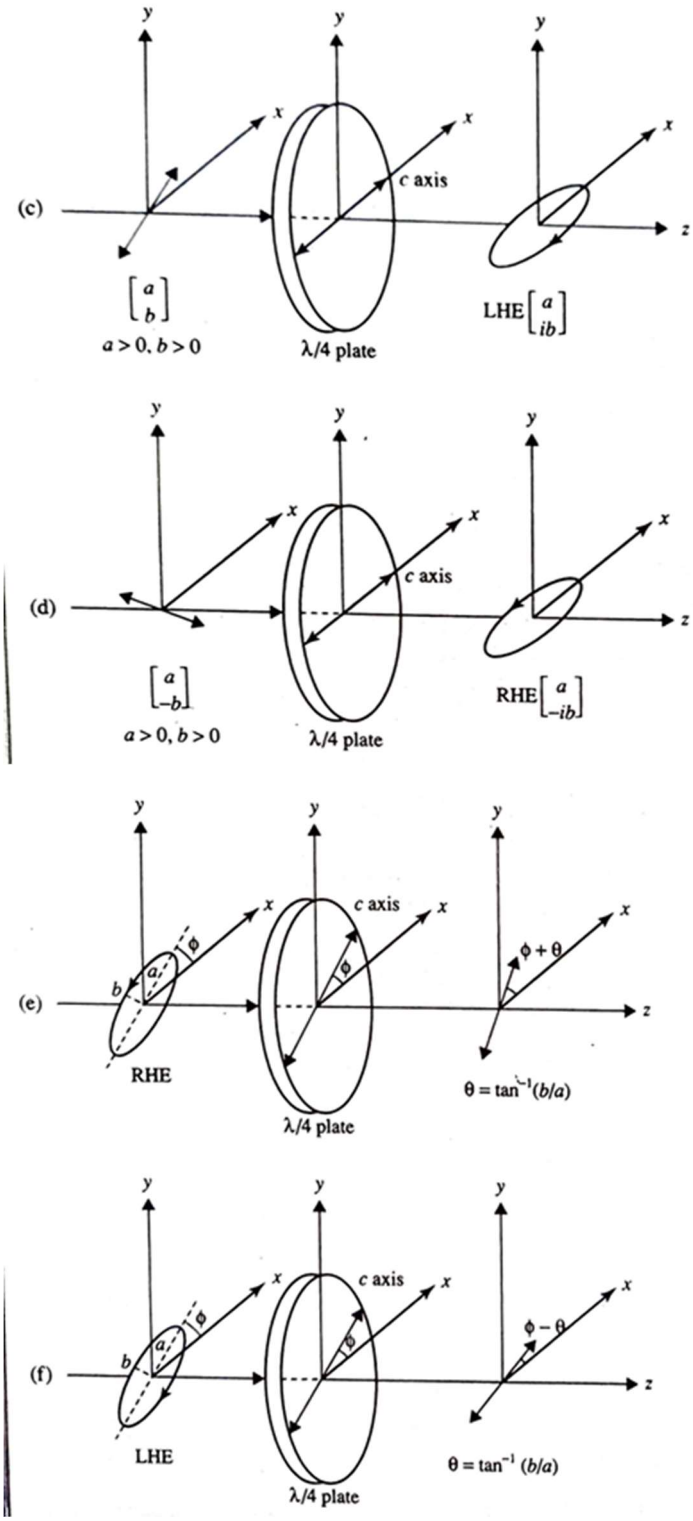


Figure 1.18 The effect of a quarter-wave plate (assuming $\Gamma = k(n_e - n_o)d = \pi/2$) on the polarization state of a linearly polarized beam. LHE = left-handed elliptically, RHE = right-handed elliptically.

TABLE 1.6 Jones Vectors

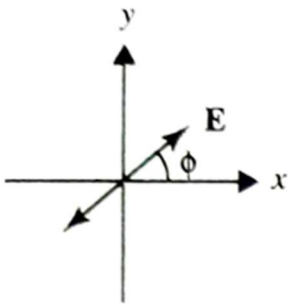
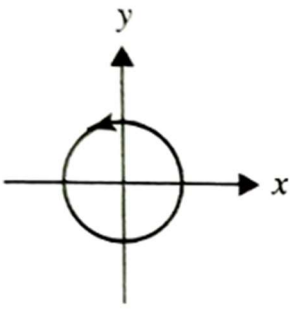
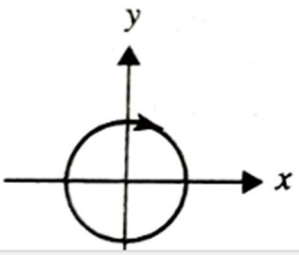
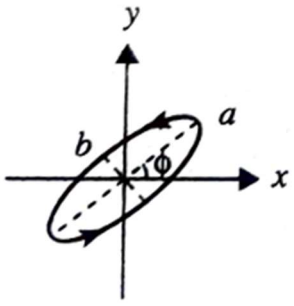
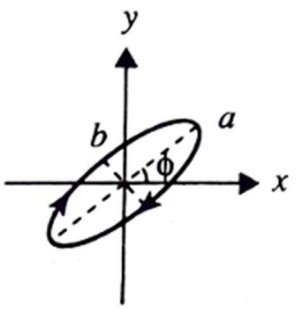
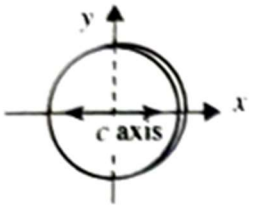
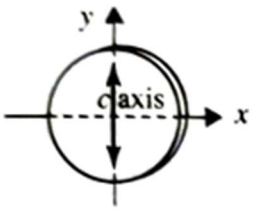
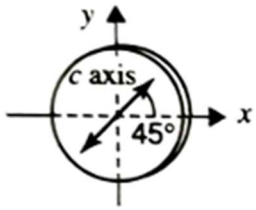
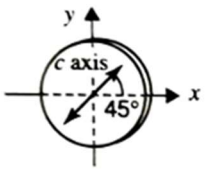
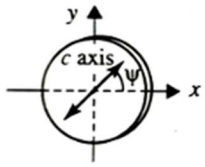
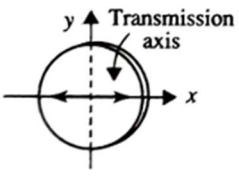
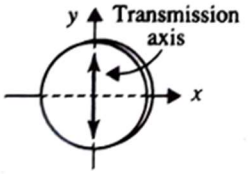
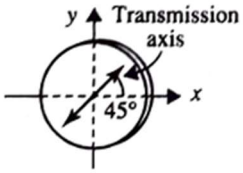
Polarization State	Jones Vector
	$\begin{bmatrix} \cos\phi \\ \sin\phi \end{bmatrix}$
	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -i \end{bmatrix}$
	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ i \end{bmatrix}$
	$\begin{bmatrix} a\cos\phi + ib\sin\phi \\ a\sin\phi - ib\cos\phi \end{bmatrix}$
	$\begin{bmatrix} a\cos\phi - ib\sin\phi \\ a\sin\phi + ib\cos\phi \end{bmatrix}$

TABLE 1.7 Jones Matrices

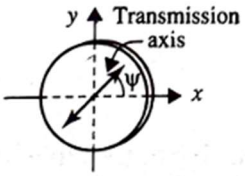
Optical Element	Jones Matrices
Wave plates	$\Gamma \equiv \frac{2\pi}{\lambda}(n_e - n_o)d$
	$\begin{bmatrix} e^{-i\Gamma/2} & 0 \\ 0 & e^{i\Gamma/2} \end{bmatrix}$
	$\begin{bmatrix} e^{i\Gamma/2} & 0 \\ 0 & e^{-i\Gamma/2} \end{bmatrix}$
	$\begin{bmatrix} \cos(\Gamma/2) & -i\sin(\Gamma/2) \\ -i\sin(\Gamma/2) & \cos(\Gamma/2) \end{bmatrix}$
	$\begin{bmatrix} \cos(\Gamma/2) & -i\sin(\Gamma/2) \\ -i\sin(\Gamma/2) & \cos(\Gamma/2) \end{bmatrix}$
	$R(-\psi)\begin{bmatrix} e^{-i\Gamma/2} & 0 \\ 0 & e^{i\Gamma/2} \end{bmatrix}R(\psi)$ $= \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}\begin{bmatrix} e^{-i\Gamma/2} & 0 \\ 0 & e^{i\Gamma/2} \end{bmatrix}\begin{bmatrix} \cos\psi & \sin\psi \\ -\sin\psi & \cos\psi \end{bmatrix}$ $= \begin{bmatrix} e^{-i\Gamma/2}\cos^2\psi + e^{i\Gamma/2}\sin^2\psi & -i\sin(\Gamma/2)\sin(2\psi) \\ -i\sin(\Gamma/2)\sin(2\psi) & e^{-i\Gamma/2}\sin^2\psi + e^{i\Gamma/2}\cos^2\psi \end{bmatrix}$
Polarizers	
	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$



$$R(-\psi)\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}R(\psi)$$

$$= \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \cos\psi & \sin\psi \\ -\sin\psi & \cos\psi \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}\begin{bmatrix} \cos\psi & \sin\psi \\ 0 & 0 \end{bmatrix} = \begin{vmatrix} \cos^2\psi & \cos\psi\sin\psi \\ \sin\psi\cos\psi & \sin^2\psi \end{vmatrix}$$